LM Physics of Climate

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10 Introduction to fluid dynamics

Climate is strongly dependent on how air moves in the atmosphere, and how water moves in the ocean. Fluid dynamics is therefore an important aspect of the physics of climate.

10.1 Going with the flow: the Lagrangian derivative

When constructing equations of motion and continuity later on, we will need to use the derivatives of certain fluid properties with respect to time. The way we are used to doing this is by using what is known as an *Eulerian* derivative, $\frac{\partial}{\partial t}$, i.e. the derivative with respect to time at a specific location. However, in an atmosphere or ocean, parcels of fluid move around. It can therefore be useful to consider derivatives within a moving parcel.

To do this, we need to account for not only changes with respect to time, but also the fact that the parcel is moving. Consider a property of the fluid, C. The change in C within a fluid parcel can be described as:

$$\begin{aligned}\delta C &= \frac{\partial C}{\partial t}\delta t + \frac{\partial C}{\partial x}\delta x + \frac{\partial C}{\partial y}\delta y + \frac{\partial C}{\partial z}\delta z \\ &= \frac{\partial C}{\partial t}\delta t + \left(u_x\frac{\partial C}{\partial x} + u_y\frac{\partial C}{\partial y} + u_z\frac{\partial C}{\partial z}\right)\delta t,\end{aligned}$$

where u_x, u_y and u_z are the fluid's velocity in the x, y and z directions, respectively, and δx, δy and δz are the distances travelled in each direction by the fluid parcel in time δt.

Dividing both sides by δt and taking the limit of infinitesimal variations, we get:

$$\frac{DC}{Dt} = \frac{\partial C}{\partial t} + \mathbf{u}\cdot\nabla C, \tag{10.1}$$

where $\mathbf{u}$ is the vector velocity of the fluid. $\frac{D}{Dt}$ is known as the *Lagrangian* or *material* derivative, and represents the rate of change of a property as you move along with a fluid parcel.

We can think of the Eulerian versus Lagrangian derivatives by picturing a river flowing under a bridge. If we stand on the bridge and observe changes in the water flowing past us, this is equivalent to the Eulerian derivative. Conversely, if we flow along with the river on a boat, we experience changes in the parcel of water which we travel with, equivalent to the Lagrangian derivative.

10.2 Mass conservation

In what follows, we will be assuming that mass is conserved in our fluid. Mathematically, the rate of change of mass within a fluid parcel of dimensions δx by δy by δz and density ρ is:

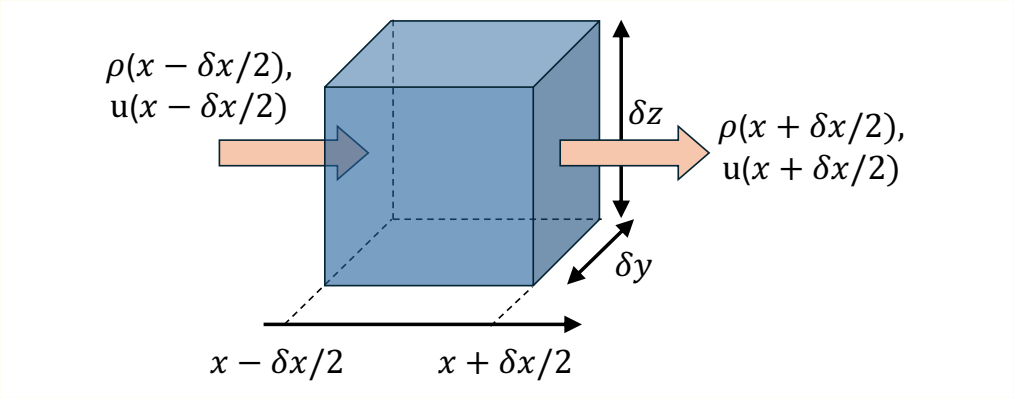


Figure 1: Schematic of a fluid parcel. The arrows denote fluid flow in the x-direction with density, ρ, and velocity, u, which change as a function of x.

$$\frac{\partial M}{\partial t} = \frac{\partial}{\partial t}\left(\rho \delta x \delta y \delta z\right) = \frac{\partial \rho}{\partial t}\delta x \delta y \delta z \tag{10.2}$$

For mass to be conserved, $\frac{\partial M}{\partial t}$ must equal the net flux of mass entering the fluid parcel. Consider the fluid parcel in Figure 1. In the x-direction, the mass flux entering the parcel on the left is $\rho(x-\delta x/2)u(x-\delta x/2)\delta y \delta z$. On the right side, the mass flux leaving the parcel is $\rho(x+\delta x/2)u(x+\delta x/2)\delta y \delta z$. Using a Taylor expansion, these fluxes can be written as $\left(\rho(x)u(x) \pm \frac{\partial(\rho u)}{\partial x}\delta x/2\right)\delta y \delta z$. The net flux of mass into the parcel in the x-direction is therefore:

$$\text{net mass flux in } x\text{−direction} = -\frac{\partial(\rho u)}{\partial x}\delta x \delta y \delta z$$

The mass fluxes in the y and z directions follow the same form. The overall mass flux into the fluid parcel is therefore

$$\text{net mass flux into parcel} = -\nabla \cdot (\rho \mathbf{u})\delta x \delta y \delta z.$$

Equating this to Equation 10.2, mass conservation can be described by:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \tag{10.3}$$

Using the Lagrangian derivative (Equation 10.1), this can equivalently be written as

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0. \tag{10.4}$$